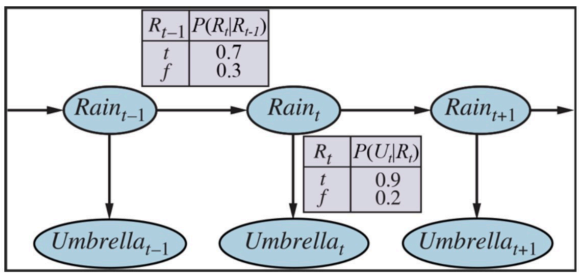
Exercise 1

Describe the *umbrella world* as an HMM:

- What is the set of unobserved variable(s) for a given time-slice t (denoted $\mathbf{X}_t$ in the book)?
- What is the set of observable variable(s) for a given time-slice t (denoted $\mathbf{E}_t$ in the book)?
- Present the *dynamic model* $\mathbf{P}(\mathbf{X}_t|\mathbf{X}_{t-1})$ and the *observation model* $\mathbf{P}(\mathbf{E}_t|\mathbf{X}_t)$ as matrices.
- Which assumptions are encoded in this model? Are the assumptions reasonable for this particular domain? (See 14.1 Time and Uncertainty on Page 479).

Figure 14.2



Bayesian network structure and conditional distributions describing the umbrella world. The transition model is $\mathbf{P}(Rain_t|Rain_{t-1})$ and the sensor model is $\mathbf{P}(Umbrella_t|Rain_t)$.

- Unobserved variable is rain
- Observable variable is umbrella

Here the dynamic model is $P(R_t|R_{t-1})$ and the observation model is $P(U_t|R_t)$.
When presented as matrices

$$P(R_t|R_{t-1})$$

where $P(R_t = t|R_{t-1} = t) = 0,7$
$P(R_t = t|R_{t-1} = f) = 0,3$
$P(R_t = f|R_{t-1} = t) = 0,3$
$P(R_t = f|R_{t-1} = f) = 0,7$

$$P(R_t|R_{t-1}) \rightarrow \begin{pmatrix} 0,7 & 0,3 \\ 0,3 & 0,7 \end{pmatrix}$$

the rows represent R_{t-1} (previous state)
the colums represent R_t (current state)

$P(U_t | R_t)$

$P(U_t = t | R_t = t) = 0,9$
$P(U_t = t | R_t = f) = 0,1$
$P(U_t = f | R_t = t) = 0,2$
$P(U_t = f | R_t = f) = 0,8$

$$P(U_t | R_t) \rightarrow \begin{pmatrix} 0,9 & 0,1 \\ 0,2 & 0,8 \end{pmatrix}$$

the rows represent unobservable variable
the colums represent observable variable

Assumptions from the model

1) first order Markov process, i.e present state is only dependent of the previous state

2) sensor independence $P(U_t | R_t, R_{t-1}, ..,) = P(U_t | R_t)$, todays observation depends only on weither it rains today

3) stationary: transiton and observation probabilities are the same for all t.

Are assumptions reasonable?

1) logically yes, but depends on where the situation takes place, i.e a place where it often rains for a longer period of time

2) reasonable if the umbrella is used only for rain today

3) not realistic for long term, e.g. time of year can affect the weather
realistic for short term, e.g. a week.

Exercise 2 on .ipynb file